AF-LeWM-lite on Official PushT

Reliable Rerun Summary

April 22, 2026

Bottom line. The reliable rerun reports **9/100** successes for **AF-LeWM-lite v1**. All models remain weak under the short matched budget, and the Wilson intervals overlap, so the current evidence supports an exploratory ranking rather than a benchmark-level claim.

Architecture and Design Principle

AF-LeWM-lite keeps the LeWM planning contract intact. Pixels are encoded by a shared ViT-tiny backbone; the dynamics projection head produces `emb`, action blocks are encoded separately, and the autoregressive predictor rolls the dynamics latent forward under candidate actions. Planning uses terminal latent distance between predicted dynamics and the goal dynamics latent.

The AF extension adds an appearance projection head `app_emb` used only for representation shaping during training. v1 adds dynamics invariance under appearance augmentation and a cross-covariance penalty between `emb` and `app_emb`. v2 adds sequence-consistent augmentation parameters, stop-gradient invariance, an appearance nuisance head, and a gradient-reversal nuisance head on the dynamics branch.

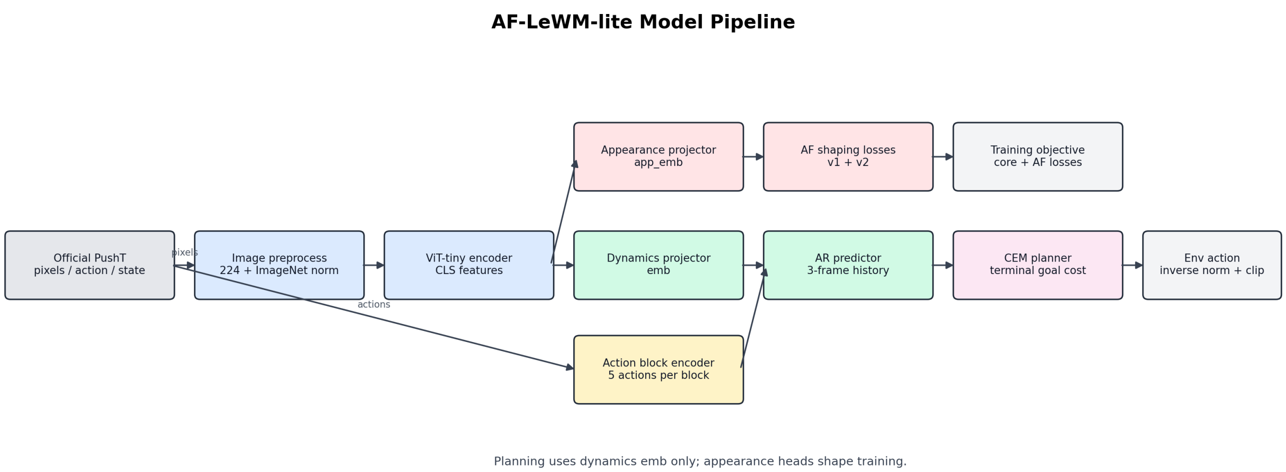


Figure 1: Model pipeline. The planner sees only the dynamics branch; appearance heads affect training losses.

Reliable Experiment Protocol

The rerun uses the official `pusht_expert_train.h5` dataset, an episode-disjoint 90/10 train/validation split, normalizers fitted only on training episodes, and fresh-run guards that refuse to mix stale checkpoint or metrics artifacts with a new run. Each model is trained for 10 epochs with 200 training batches per epoch, 20 validation batches per epoch, batch size 4, bf16 precision, and seed 3072.

Each epoch-10 object checkpoint is evaluated on two independent 50-start samples, `seed=42` and `seed=43`.

The report aggregates 100 trials per model and reports Wilson 95% intervals. Cross-model loss comparisons use the shared JEPA core objective $\mathcal{L}_{core} = \mathcal{L}_{pred} + 0.09\mathcal{L}_{sigreg}$.

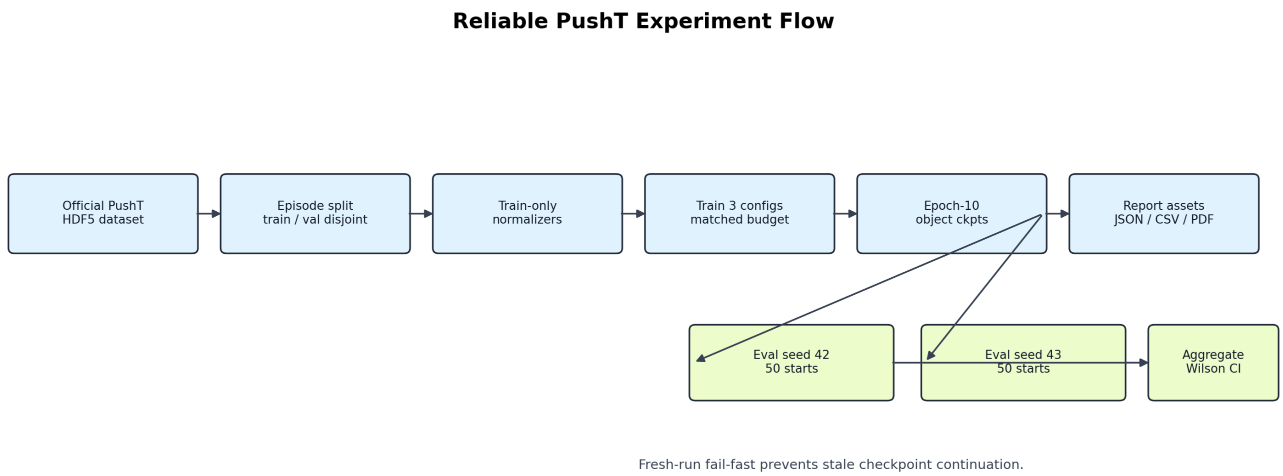


Figure 2: Experiment flow from official dataset to two-seed aggregate report.

Implementation Fixes

- Training now uses episode-disjoint subsets rather than overlapping clip-level random split.
- Action, proprioception, and state normalizers are fitted on training episodes only.
- Fresh reliable runs fail fast when old metrics or checkpoints already exist.
- Evaluation reconstructs the raw 15-step history window and compresses it to the model's 3-frame context.
- Goal indexing is asserted against the planned raw-step horizon.
- CEM candidate actions, elite means, and returned normalized plans are clamped to normalized env action bounds before inverse transform.
- Evaluation writes row, episode, and start-step provenance for every sampled rollout.

Results

Model	Params (M)	Train (min)	Core loss	Seed 42	Seed 43	Aggregate
Baseline LeWM	18.03	9.69	0.14731	2/50	3/50	5/100 = 5.0%
AF-LeWM-lite v1	18.17	13.30	0.15033	4/50	5/50	9/100 = 9.0%
AF-LeWM-lite v2	18.24	14.02	0.15139	1/50	5/50	6/100 = 6.0%

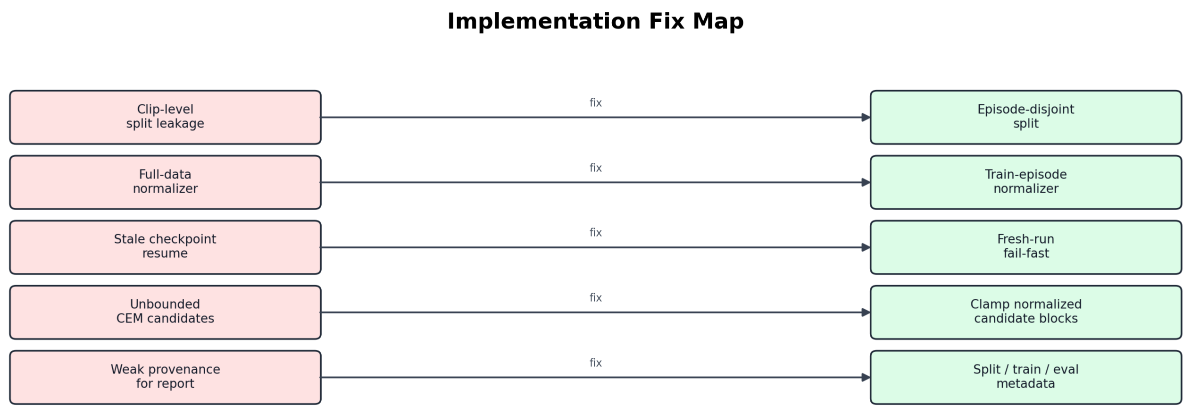


Figure 3: Reliability fixes that directly affect whether the experiment tests the intended design.

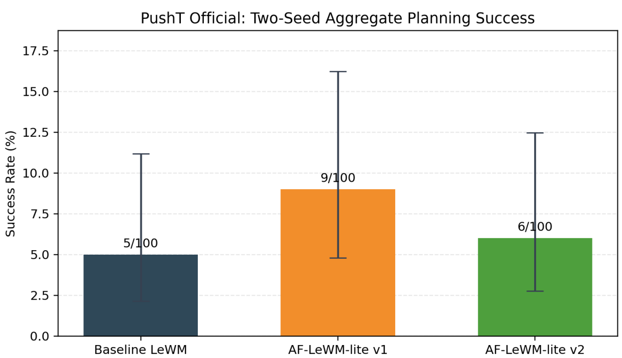
Model	Aggregate success	Wilson 95% CI
Baseline LeWM	5/100 = 5.0%	[2.15%, 11.18%]
AF-LeWM-lite v1	9/100 = 9.0%	[4.81%, 16.23%]
AF-LeWM-lite v2	6/100 = 6.0%	[2.78%, 12.48%]

Interpretation

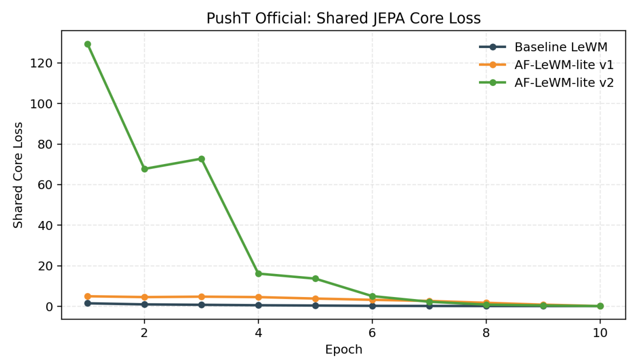
The corrected result should be read conservatively. The best model in this rerun is **AF-LeWM-lite v1**, but the aggregate success counts are small and the intervals overlap. The current study is useful as a reliability-checked ablation of the AF shaping idea; it does not establish a strong PushT control result.

The most important conclusion is methodological: after removing split leakage, full-data normalization, stale checkpoint continuation, and invalid CEM candidate actions, any remaining weakness is visible in the audited protocol rather than hidden in the implementation.

Generated from `tools/generate_pusht_official_report_assets.py` using live run metrics, split metadata, train metadata, and evaluation result files under `STABLEWM_HOME`.



(a) Two-seed planning success with Wilson intervals.



(b) Shared JEPA core loss across training epochs.